

Checkpoint

12.4 The Cross Product

1 Problems

Problem 1. Find $\langle 1, 2, 3 \rangle \times \langle 3, 2, 1 \rangle$.

Problem 2. What is the difference between $\vec{u} \times \vec{v}$ and $\vec{v} \times \vec{u}$?

Problem 3. Geometrically, what is $2|\vec{u} \times \vec{v}|$?

Problem 4. If $\vec{u} = \hat{i} - 2\hat{j} + \hat{k}$, find $\vec{u} \times \vec{u}$. Does the specifics of $\vec{u}$ even matter in this case?

Problem 5. If The angle between $\vec{u}$ and $\vec{v}$ is 45 degrees, and $|\vec{u}| = 2$ and $|\vec{v}| = 3$, then what is $|\vec{u} \times \vec{v}|$?

Problem 6 (**Most Important Problem**). What does it mean for $\vec{x}$ to be the cross product of $\vec{u}$ and $\vec{v}$?

2 Answers

1. $\langle -4, 8, -4 \rangle$
2. One is the negative of the other, that is $\vec{u} \times \vec{v} = -(\vec{v} \times \vec{u})$
3. it is twice the area of the parallelogram formed by the vectors $\vec{u}$ and $\vec{v}$.
4. $\langle 0, 0, 0 \rangle$ and no, $\vec{u} \times \vec{u} = \langle 0, 0, 0 \rangle$ for any $\vec{u}$.
5. $3\sqrt{2}$
6. $\vec{x}$ is orthogonal to both $\vec{u}$ and $\vec{v}$